

1. Productivity growth—Real per capita GDP in the United States has increased at an average rate of near 2 percent over the past 100 years as a result of productivity gains, but has fluctuated widely around this trend based on the prevailing long-term and business cycles.

2. Long-term credit expansion/deleveraging cycle—Initially, the availability of credit expands spending beyond income levels. As Dalio explains,

[This process] is self-reinforcing because rising spending generates rising incomes and rising net worths, which raise borrowers' capacity to borrow, which allows more buying and spending. . . . The up-wave in the cycle typically goes on for decades, with variations in it primarily due to central banks tightening and easing credit (which makes business cycles).

Although self-reinforcing, the credit expansion phase ultimately reaches a point where it can no longer be extended. Dalio describes this transition in the credit cycle as follows:

It can't go on forever. Eventually the debt service payments become equal to or larger than the amount we can borrow and the spending must decline. When promises to deliver money (debt) can't rise any more relative to the money and credit coming in, the process works in reverse and we have deleveragings. Since borrowing is simply a way of pulling spending forward, the person spending \$110,000 per year and earning \$100,000 per year has to cut his spending to \$90,000 for as many years as he spent \$110,000, all else being equal. . . . In deleveragings, rather than debts rising relative to money as they do in up-waves, the reverse is true. As the money coming in to debtors via incomes and borrowings is not enough to meet debtors' obligations, assets need to be sold and spending needs to be cut in order to raise cash. This leads asset values to fall, which reduces the value of collateral, and in turn reduces incomes. Because of both lower collateral values and lower incomes, borrowers' creditworthiness is reduced, so they justifiably get less credit, and so it continues in a self-reinforcing manner.